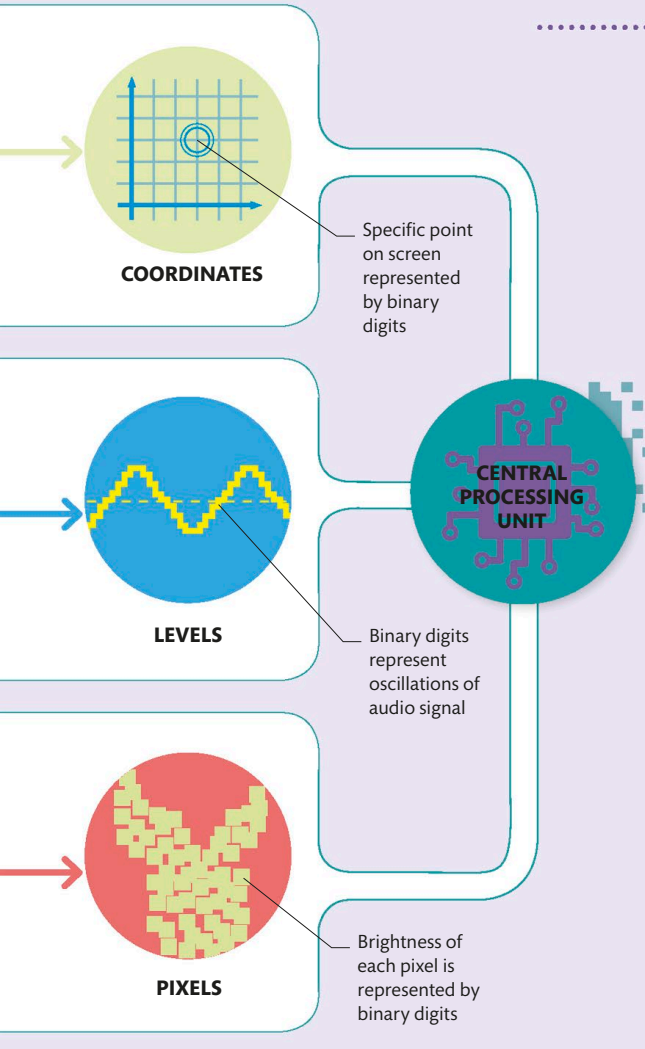
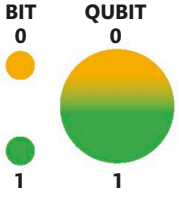


Digital signals
All the different ways in which information is digitized produce large collections of binary numbers that are processed by the central processing unit (CPU) of a computer embedded inside a digital device.

BASE 10 (DECIMAL)			BASE 2 (BINARY)		
12	4	7	1100	100	111
8	16	2	1000	10000	10
20	5	15	10100	101	1111
9	17	21	1001	10001	10101

QUANTUM COMPUTING

All digital devices currently available use bits, which can take on only one value at a time, and contain computers that process instructions one at a time. Computer scientists and physicists are developing quantum computers that will use qubits, which can take on many values at the same time. By combining qubits, computers will be able to carry out a potentially infinite number of instructions, promising much faster devices in the future.



WHAT ARE DATA?

Data (singular datum) are pieces of information. In the digital world, data refers to any information stored and processed inside digital devices. It includes personal information about the users of digital devices.

UNITS OF DIGITAL INFORMATION

Unit	Size	Application
Byte (B)	8 bits	A basic unit of information held by computers, a byte is equivalent to eight bits (binary digits).
Kilobyte (KB)	1,000 bytes	A short, simple text file on a computer would take up a few kilobytes.
Megabyte (MB)	1 million bytes	One million bytes (8 million bits) could represent one minute of digital sound.
Gigabyte (GM)	1 billion bytes	One billion bytes (8 billion bits) could represent 4,000 digital images.
Terabyte (TB)	1 trillion bytes	Computer hard disks are often this size, able to store large amounts of digital information.